

Phetolo Tshukudu

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GitHub: github.com/phetolo-tshukudu

PROFESSIONAL SUMMARY

Final-year BSc Mathematical Sciences student at the University of Johannesburg, majoring in Computer Science and Mathematical Statistics, with a strong foundation in software engineering, backend development and data analysis. Hands-on experience building Java/Spring Boot REST APIs and Python data-analysis applications, with practical experience in relational databases, object-oriented programming, automated testing and statistical analysis. Interested in graduate and junior software engineering, backend development and data engineering opportunities where I can apply programming, analytical and problem-solving skills to build reliable, data-driven systems.

EDUCATION

University of Johannesburg | 2023 - Present

BSc Mathematical Sciences in Computer Science and Mathematical Statistics

Relevant coursework: Object-Oriented Programming; Data Structures & Algorithms; Database Systems; Software Engineering; Computer Networks; Operating Systems; Mathematical Statistics; Statistical Modelling; Regression Analysis.

Schooling / Career Highlights

- High school learner council
- Academic award in Mathematics
- UJ UJenius Certificate

SKILLS

- **Languages:** Java, Python, SQL, R
- **Backend:** Spring Boot, Spring Security, REST APIs, JWT, JPA/Hibernate
- **Python & Data:** Pandas, data cleaning, data analysis, statistical analysis, dataclasses
- **Databases:** MySQL, SQLite, relational database design
- **Testing:** Pytest, JUnit 5, Mockito
- **Development Tools:** Git, GitHub, Maven, Eclipse IDE, Postman, MySQL Workbench
- **Core Computer Science:** Data structures and algorithms, graph algorithms, database systems, software engineering
- **Statistics & Data:** Statistical modelling, regression analysis, data interpretation and visualisation, model evaluation and validation

WORK EXPERIENCE (PART TIME)

Mathematics & Physical Sciences Tutor

Ghetto Hero Saturday School | Jun 2023 - Jul 2023

- Tutored Grade 10-12 learners in Mathematics and Physical Sciences.
- Explained complex concepts using structured, accessible problem-solving approaches.
- Supported learners in developing mathematical reasoning and problem-solving skills.

LEADERSHIP EXPERIENCE

Team Leader - Free Sanitary Pad & Toiletry Project

Bridge to Employment Programme, NICDAM

- Led a team in organising a community project providing sanitary products and toiletries.
- Coordinated team responsibilities and project activities.
- Engaged potential sponsors to seek donations and support for the initiative.
- Developed leadership, teamwork, communication and project coordination skills.

SELECTED PROJECTS

COMMUNITY INFRASTRUCTURE REPORTING API

Java 21 | Spring Boot | Spring Security | JWT | BCrypt | MySQL | Flyway | Docker

- Built a secure API where residents report infrastructure issues and workers/admins review incidents through verification, work and resolution.
- Implemented role-based access, coordinate validation, filtering and analytics; containerised the API and MySQL with Docker Compose, health checks and persistent storage.
- Recorded the author and timestamp of status changes; used optimistic locking to reject conflicting updates and roll back associated history.

github.com/phetolo-tshukudu/infrastructure-reporting

FINCORE - PERSONAL FINANCE BACKEND

Java 21 | Spring Boot | JWT | PostgreSQL | Flyway | Docker | OpenAPI

- Built a REST API for managing users, accounts, transactions and personal budgets.
- Implemented authentication, persistence, schema migrations, validation, API documentation and Docker support.

github.com/phetolo-tshukudu/fincore

FINANCIAL DATA ANALYZER

Python | Pandas | Pytest | Dataclasses | Statistics

- Built an application for cleaning, processing and analysing financial transaction data.
- Added spending statistics, budget tracking and 16 automated pytest tests for core business logic.

PERSONALFINANCE API - FINANCIAL MANAGEMENT BACKEND

Java | Spring Boot | Spring Security | JWT | MySQL | JPA/Hibernate | JUnit 5 | Mockito

- Developed a layered REST API for users, accounts, transactions and budgets.
- Implemented security, persistence, validation, exception handling and unit tests.

github.com/phetolo-tshukudu/finance-api

WILDFIRE SIMULATOR

Java | JavaFX | Graph Algorithms | Dijkstra's Algorithm

- Collaborated on an application that simulates wildfire spread and identifies safer escape routes.
- Used Dijkstra's algorithm, priority queues and custom data structures to model routes and fire conditions.

github.com/phetolo-tshukudu/Wildfire-Simulator